

Natural Language Processing – DS4503

Student Name	Student ID
Eyad Mohammed Al-harhi	444000005
Mohammed Nedal Al-sharif	444004824
Ziyad Omer Al-talhi	444000039
Yazan Ibrahim Al-Ghamdi	444000120
Ayman Abdulghani Al-Zahrani	444006406

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1. Introduction

In this project, Natural Language Processing (NLP) techniques were applied to analyze text data. The main focus of the project is to perform **sentiment analysis** by classifying tourist reviews into positive or negative sentiment. The project follows the text classification track and applies the methods learned during the course.

2. Project Idea and Problem

Tourist attractions receive a large number of written reviews from visitors. Reading and analyzing these reviews manually takes a lot of time and effort. The idea of this project is to use NLP techniques to automatically classify reviews based on sentiment, which helps summarize visitor opinions in a simple and efficient way.

3. Dataset Description

The dataset consists of multiple data files, where each file contains tourist reviews from a specific cultural or heritage location in Saudi Arabia. The data includes textual reviews and rating values. All reviews are written by real users, which makes the dataset suitable for sentiment analysis.

4. Data Preparation

In the data preparation stage, all datasets were loaded and cleaned. Empty and missing reviews were removed to improve data quality. Text cleaning and normalization were applied to prepare the reviews for feature extraction and model training.

5. Sentiment Labeling

The rating values were converted into sentiment labels as follows:

- **Negative:** reviews with ratings of 1–2 stars
- **Positive:** reviews with ratings of 3–5 stars

This step transforms numerical ratings into sentiment categories that can be used for supervised learning.

6. Feature Representation

To convert text into numerical features, the **TF-IDF (Term Frequency–Inverse Document Frequency)** method was used. This method represents reviews based on the importance of words in the dataset and is commonly used in text classification tasks.

7. Model Training

The dataset was split into training and testing sets. A classification model was trained using the TF-IDF features to learn how to distinguish between positive and negative reviews.

8. Model Evaluation

The model performance was evaluated using standard metrics such as **Accuracy** and **F1-score**. These metrics help measure how well the model performs on unseen data.

9. Results and Discussion

The results show that the model is able to classify tourist reviews into positive and negative sentiment. The analysis provides a general understanding of visitor opinions across different tourist locations and shows the usefulness of NLP techniques in analyzing real-world text data.

10. Conclusion

This project successfully implements a complete NLP pipeline, starting from data preparation and sentiment labeling to feature extraction, model training, and evaluation. The project demonstrates how sentiment analysis can be applied to tourist reviews to extract meaningful insights in a clear and simple way.